

# THE UNIVERSITY OF HONG KONG UNOFFICIAL MOCK

## PHYS1150: PROBLEM SOLVING IN PHYSICS

Answer ALL questions and put your answers on the answer book provided.

This examination paper has 12 pages (including this front page and two pages of list of formulae at the end). There are 6 problems. Answer ALL of them.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of their examination script.

**Problem 1: Short Answer Question (20 marks)**

For question 1.1 and 1.2, there are some numbered choices which one or more of them may be correct. Sum up all the numbers of correct choices and write the result in the box on the answer paper. If none of the choices are correct, answer 0. No explanation is required. For question 1.3 and 1.4, put down answer only. No explanation is required.

**1.1 (5 marks)**

Which of the following statements is/are true?

- (1) For any twice differentiable  $f(x)$  where  $f(0) = f'(0) = f''(0) = 6$ . Then  $\lim_{x \rightarrow 0} \frac{f(\sin^2 x)}{x^4} = 3$ .
- (2)  $\lim_{n \rightarrow \infty} \sin^2(\pi\sqrt{n^2 + n})$  does not exist.
- (4) For any multi-variable function  $z = z(x, y)$  where  $F(x + \frac{z}{y}, y + \frac{z}{x}) = 0$  for some functions  $F$ ,

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z - xy.$$

- (8) For  $y = y(x)$  satisfying  $e^{3y} + \int_0^{x+y} \cos(t^2) dt = 1$ , at  $(0, 0)$ ,  $y'' = \frac{9}{64}$ .

**1.2 (5 marks)**

Which of the following statements is/are true?

- (1) If  $A$  has an eigenvalue of  $\lambda$ , then  $A^T$  has an eigenvalue of  $\lambda$ .
- (2) If  $A^T$  has an eigenvalue of  $\lambda$ , then  $A$  has an eigenvalue of  $\lambda$ .
- (4) If  $A$  has an eigenvector of  $\vec{v}$ , then  $A^T$  has an eigenvector of  $\vec{v}$ .
- (8) If  $A^T$  has an eigenvector of  $\vec{v}$ , then  $A$  has an eigenvector of  $\vec{v}$ .

**1.3 (5 marks)**

Find  $f(x)$  such that  $f(x) \cos(x) + 2 \int_0^x f(t) \sin(t) dt = x + 1$ .

**1.4 (5 marks)**

Approximate  $\int_0^1 \frac{e^{\sin(x)} \sin(2x)}{(1 - \sin x)^2} dx$  using trapezium rule with 5 subintervals.

**Problem 2: Laplace Transform (20 marks)**

Let  $f(t)$  be a function defined for  $t \geq 0$ . The *Laplace transform* of  $f$ , denoted by  $\mathcal{L}\{f(t)\}$  or  $F(s)$ , is defined as

$$\mathcal{L}\{f(t)\}(s) = F(s) = \int_0^{\infty} e^{-st} f(t) dt,$$

for all real numbers  $s$  for which the improper integral converges. (You may neglect the condition of convergence in this problem.)

**(a)** Let  $Y(s) = \mathcal{L}\{y(t)\}$  where  $\mathcal{L}\{y(t)\}$  exists.

Prove that

$$\mathcal{L}\{y''(t)\} = sY(s) - y(0)$$

and

$$\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0).$$

**(4 marks)**

**(b)** Find  $\mathcal{L}\{\sin t\}(s)$ . **(3 marks)**

**(c)** Let  $y(t)$  be the solution of the linear constant-coefficient differential equation

$$a_n y^{(n)}(t) + a_{n-1} y^{(n-1)}(t) + \cdots + a_1 y'(t) + a_0 y(t) = f(t),$$

with initial conditions

$$y(0) = y_0, \quad y'(0) = y_1, \quad \dots, \quad y^{(n-1)}(0) = y_{n-1}.$$

Assume that  $y(t)$  and its derivatives are of exponential order so that their Laplace transforms exist. Let  $Y(s) = \mathcal{L}\{y(t)\}$ . Then apply the Laplace transform to the expression on the both sides and solve for  $Y(s)$ . Then  $y(t) = \mathcal{L}^{-1}\{Y(s)\}$ .

Given that  $\mathcal{L}^{-1}\{1/(s+a)\} = e^{-at}$ ,  $\mathcal{L}^{-1}\{s/(s^2+1)\} = \cos(t)$ .

Solve  $y'' + 3y' + 2y = \sin(t)$  where  $y(0) = 1$  and  $y'(0) = 0$ . **(7 marks)**

**We use another way to verify the answer in (c).**

**(d)** Solve for general solution of  $y'' + 3y' + 2y = 0$ . Steps are optional. **(2 marks)**

**(e)** The solution in (c) can be written as  $y_c(x) + y_h(x)$  where  $y_c$  is the solution in (d) and

$$y_h = A \sin(t) + B \cos(t)$$

where  $A$  and  $B$  are constants to be determined. Verify the answer in (c). **(4 marks)**

△ The use of Laplace transform is the primary method for analyzing the switch-on/switch-off behavior (transient response) of RLC circuits and control systems.

### Problem 3: Tensor (15 marks)

Let  $M$  be the set of all  $3 \times 1$  real matrices. For any  $u, x \in M$ , we define their tensor (outer) product

$$u \otimes x = ux^T,$$

which is a  $3 \times 3$  real matrix.

(a) Show that for any  $u, x, y \in M$  and  $\alpha, \beta \in \mathbb{R}$ :

(i)

$$u \otimes (\alpha x + \beta y) = \alpha(u \otimes x) + \beta(u \otimes y),$$

(ii)

$$(\alpha u + \beta v) \otimes x = \alpha(u \otimes x) + \beta(v \otimes x) \quad \text{for any } u, v \in M.$$

(b) Show that if  $u \otimes x = 0$  for all  $x \in M$ , then  $u = 0$ . Deduce that if  $u \otimes x = v \otimes x$  for all  $x \in M$ , then  $u = v$ .

(c) Let  $A$  be a  $3 \times 3$  real matrix and  $u \in M$  such that the  $i$ -th column of  $A$  equals the  $i$ -th column of  $u \otimes e_i$  for each standard basis vector  $e_i$ :

$$Ae_i = (u \otimes e_i) \mathbf{e}_i \quad \text{for } i = 1, 2, 3,$$

(where  $\mathbf{e}_i$  extracts the  $i$ -th column). Show that this condition forces all columns of  $A$  to be equal to  $u$ .

(d) With  $u = \begin{pmatrix} p \\ q \\ r \end{pmatrix}$ , determine the matrix  $A$  whose three columns are all equal to  $u$ . Hence compute  $Ax$  explicitly for a general vector  $x \in M$ .

⚠ In the real world, many physical properties are direction-dependent. For instance, the stress inside a structural beam is not just one number; it depends on both the surface area you are looking at and the direction of the force applied. Tensors are the necessary mathematical generalization of scalars (rank-0) and vectors (rank-1) to describe these complex, multi-directional quantities. They are crucial for fields ranging from geology (describing crystal properties) to General Relativity (where the stress-energy tensor describes how matter and energy warp spacetime).

## Problem 4: Lie Algebra (5 marks)

A *Lie algebra* over a field  $\mathbb{F}$  is a pair  $(\mathfrak{g}, [\cdot, \cdot])$  consisting of:

1. a vector space  $\mathfrak{g}$  over  $\mathbb{F}$ ,
2. a bilinear map (called the *Lie bracket*)

$$[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g},$$

such that for all  $x, y, z \in \mathfrak{g}$  and all scalars  $\alpha, \beta \in \mathbb{F}$ , the following axioms hold:

$$[\alpha x + \beta y, z] = \alpha[x, z] + \beta[y, z], \quad [z, \alpha x + \beta y] = \alpha[z, x] + \beta[z, y].$$

$$[x, y] = -[y, x].$$

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

Let  $\mathfrak{g}$  be the set of all  $2 \times 2$  real matrices of the form

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}, \quad a, b, c \in \mathbb{R}.$$

- (a) Show that  $\mathfrak{g}$  is a vector space over  $\mathbb{R}$  under the usual matrix addition and scalar multiplication.
- (b) Prove that  $(\mathfrak{g}, [\cdot, \cdot])$  is a Lie algebra where the bracket is defined as follows:

$$[X, Y] = XY - YX.$$

△ Symmetry is one of the most fundamental concepts in physics, linking directly to conservation laws (e.g., time symmetry implies energy conservation). In quantum mechanics and particle physics, the rules governing these symmetries are not simple numbers, but structured collections of mathematical operations. Lie Algebras provide the framework for classifying and manipulating these symmetry groups, allowing physicists to predict the existence and properties of elementary particles and understand the fundamental forces that govern the universe.

## Problem 5: Just a Normal Linear Algebra and Calculus Problem

(20 marks)

(a) In  $\mathbb{R}^2$ , derive the rotation matrix about the origin by  $\theta$  using the concept of change of basis.

(b) For the region bounded by  $x = a$ ,  $x = b$  and  $y = f(x)$ , derive the volume of solid generated by revolving the region about  $x$ -axis by  $2\pi$ .

(c) Denote  $\mathcal{R}$  as the region bounded by  $y \geq |x|$  and  $x + y \leq \frac{\pi\sqrt{2}}{4}$ . Curve  $\mathcal{S}$  has the equation  $\left(\frac{x^2}{2} + \frac{y^2}{2} - xy\right)^2 = \tan\left[\frac{\sqrt{2}}{2}(x + y)\right]$ . Using (a) and (b), compute the volume of solid generated by revolving the region bounded by  $\mathcal{R}$  and  $\mathcal{S}$  about  $y = x$  by  $2\pi$ .

## Problem 6: Variation of Parameters from PHYS2150 and Beyond (20 marks)

(a) Throughout the question, we assume  $y_1(t), y_2(t)$  satisfy

$$y''(t) + p(t)y'(t) + q(t)y(t) = 0$$

where  $p(t), q(t)$  are continuous functions on  $I$ .

(b) Show that if  $y_1(t), y_2(t)$  are linearly independent,  $y_1(t)y_2'(t) \neq y_2(t)y_1'(t)$  for some  $t$ . Show also that if  $y_1(t)y_2'(t) \neq y_2(t)y_1'(t)$  for some  $t$ , then  $y_1(t), y_2(t)$  are linearly independent.

(c) Prove that

$$y_1(t)y_2'(t) - y_2(t)y_1'(t) = A \exp\left(-\int p(t) dt\right) \quad \text{for some constant } A.$$

(d) Define the Wronskian

$$W(y_1, y_2)(t) = y_1(t)y_2'(t) - y_2(t)y_1'(t),$$

and suppose  $g(t)$  is continuous. Let

$$Y(t) = u_1(t)y_1(t) + u_2(t)y_2(t),$$

where  $u_1(t), u_2(t)$  are smooth functions i.e. infinitely differentiable. If

$$Y''(t) + p(t)Y'(t) + q(t)Y(t) = g(t),$$

derive a system of equations for  $u_1'(t), u_2'(t)$ . Also, show that

$$Y(t) = -y_1(t) \int \frac{y_2(t)g(t)}{W(y_1, y_2)(t)} dt + y_2(t) \int \frac{y_1(t)g(t)}{W(y_1, y_2)(t)} dt.$$

(e) Verify that

$$y_1(t) = \frac{\sin t}{\sqrt{t}}$$

satisfies

$$t^2 y''(t) + t y'(t) + \left(t^2 - \frac{1}{4}\right) y(t) = 0.$$



(f) Find all  $y(t)$  satisfying

$$t^2 y''(t) + ty'(t) + \left(t^2 - \frac{1}{4}\right) y(t) = 3t^{3/2} \sin t.$$

△ The variation of parameters method is an advanced technique used to solve second-order non-homogeneous equations, particularly those modeling phenomena where an external force is applied. It is essential for accurately predicting the motion of driven harmonic oscillators, analyzing mechanical vibrations in structures, and simulating the complex, forced response of electronic filters and circuits.

# Formula Sheet

## Differentiation Formulae

| $f(x)$        | $f'(x)$                 |
|---------------|-------------------------|
| $c$           | $0$                     |
| $cx$          | $c$                     |
| $cx^n$        | $cnx^{n-1}$             |
| $\sin x$      | $\cos x$                |
| $\cos x$      | $-\sin x$               |
| $\tan x$      | $\sec^2 x$              |
| $\sec x$      | $\sec x \tan x$         |
| $\csc x$      | $-\csc x \cot x$        |
| $\cot x$      | $-\csc^2 x$             |
| $e^x$         | $e^x$                   |
| $\ln x$       | $\frac{1}{x}$           |
| $a^x$         | $a^x \ln a$             |
| $u \pm v$     | $u' \pm v'$             |
| $uv$          | $uv' + vu'$             |
| $\frac{u}{v}$ | $\frac{vu' - uv'}{v^2}$ |

**Chain Rule:**  $\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$

**Reciprocal Rule:**  $\frac{dy}{dx} = \frac{1}{\left(\frac{dx}{dy}\right)}$

## Integration Formulae

| $f(x)$                   | $\int f(x) dx$                                                                  |
|--------------------------|---------------------------------------------------------------------------------|
| $c$                      | $cx$                                                                            |
| $x$                      | $\frac{x^2}{2}$                                                                 |
| $cx^n \quad (n \neq -1)$ | $\frac{cx^{n+1}}{n+1}$                                                          |
| $\sin x$                 | $-\cos x$                                                                       |
| $\cos x$                 | $\sin x$                                                                        |
| $\sec^2 x$               | $\tan x$                                                                        |
| $\csc^2 x$               | $-\cot x$                                                                       |
| $\sec x \tan x$          | $\sec x$                                                                        |
| $\csc x \cot x$          | $-\csc x$                                                                       |
| $\sec x$                 | $\ln(\sec x + \tan x)$ or $\ln \tan \left( \frac{\pi}{4} + \frac{x}{2} \right)$ |
| $\csc x$                 | $\ln(\csc x - \cot x)$ or $\ln \tan \frac{x}{2}$                                |
| $\tan x$                 | $\ln \sec x$                                                                    |
| $\cot x$                 | $\ln \sin x$                                                                    |
| $e^x$                    | $e^x$                                                                           |
| $\frac{1}{x}$            | $\ln x$                                                                         |
| $\ln x$                  | $x \ln x - x$                                                                   |

**Integration by Parts:**  $\int u dv = uv - \int v du$

## Trigonometric Identities

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 1 - 2 \sin^2 A = 2 \cos^2 A - 1$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$\sin A + \sin B = 2 \sin \frac{A + B}{2} \cos \frac{A - B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A + B}{2} \sin \frac{A - B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A + B}{2} \cos \frac{A - B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A + B}{2} \sin \frac{A - B}{2}$$

$$\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2}[\sin(A + B) - \sin(A - B)]$$

$$\cos A \cos B = \frac{1}{2}[\cos(A + B) + \cos(A - B)]$$

$$\sin A \sin B = -\frac{1}{2}[\cos(A + B) - \cos(A - B)]$$

**END OF PAPER**